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Summary

Parameters are found. Experimental techniques required for parameter extraction are stated. LFP 26650 cell C/30 simulation is successfully completed. During 1C simulation solver error occurred.

Report

Simple battery particle stops simulation as Stoichiometry reaches 0 or 1.

Solver failed 1C scenarios. So 1C charging and discharging capacities are different which they shouldn't be.

Higher current caused activation overpotential effect more prominent. So, charging required higher terminal potential and discharging at higher current caused energy loss so obtained voltage dropped.

IR drop is also effective for voltage profile. Current collector resistance is mainly caused by contact resistance. There was no reliable data obtainable for the specific modelled cell. However a range of resistance can be considered. For the model low contact resistance is selected.

Diffusion constants were also effective at capacity loss. Because electrode surface saturates earlier at high C. Plus, overvoltage is required to drive diffusion and overcome concentration polarization.

Even if the solver failed, difference in high C charging and discharging can be visible. In modern cells anode limits charging and cathode limits discharging since intercalation has slow intercalation kinetics. But old cells like the one modelled in this work, are not nano-processed. So in either case the charge and discharge were cathode limited.

Answer to Task 1.

Parameters

To model LFP battery, parameters are selected from one resource to increase coherency between parameters. A cathode allowing higher stoichiometry may also have higher diffusion constant. For parameters other than geometry parameters, electrochemical techniques are required.

Geometry Parameters: For reverse engineering purposes thickness can be measured at SEM.

Particle radius can be found from SEM analysis. To have statistically correct particle size data, long analysis hours are required.

Max-Min Stoichiometry of electrodes can be found from optical emission spectroscopy. It is the best method to measure Li at% at different SOC. A standard oes instrument might not be sensitive to all elements, such as cobalt. So initial EDS scan is required. Plus, coin cell reconstruction from recovered electrodes can be used as reference for comparison.

Active material amount can be obtained by both combination EDS-zaf corrected and XRD analysis. OES analysis is required for best accurate results.

EIS

Kinetic and thermal data can be obtained with electrochemical techniques. The EIS analysis is a very strong tool that allows inspecting resistive, inductive or capacitive behaviors of the battery. This technique also provides information about charge and mass transfer. With use of an equivalent circuit, componentwise characterization is possible. For example, electrolyte, electrode or interface behaviors can be distinguished. To obtain the following parameters, EIS can be utilized:

- 1- Activation energy for the diffusion inside the active material of the electrode
- 2- Activation energy for the diffusion inside the electrolyte
- 3- Activation energy for the charge transfer inside the electrode
- 4- Activation energy for the conduction inside the electrode and electrolyte
- 5- Conductivity of components
- 6- Resistance of current collector such as contact resistance
- 7- Diffusion coefficient of the active material in the electrode
- 8- Rate constant of the charge transfer for the electrode
- 9- Bruggeman exponent of the separator

Answer to Task 2.

A123 ANR26650 M1 LFP cell cited in [Simplified Electrochemical and Thermal Model of LiFePO₄-Graphite Li-Ion Batteries for Fast Charge Applications](#) is selected. Parameters are listed in [meatec parameters.gsheel](#).

NON EIS

- 1- Bruggeman exponent of the electrodes

Tortuosity measurement is required. So CT scanning is required.

Using EIS ionic conduction resistance of comparative measurements of bulk electrode and known porosity. Potentially high error.

- 2- Fraction of the total electric current that an ion carries in an electrolyte

Steady-state concentration polarization measurement (Bruce–Vincen, Hitorff methods)

- 3-Electrolyte phase Li⁺ diffusion coefficient

GITT

TOGETHER WITH EIS

- 1- Rate constant of the charge transfer for the electrode

CV

2- Activation energy for the diffusion inside the active material of the electrode

GITT

3- Activation energy for the conduction inside the electrode and electrolyte

Current Interrupt

Answer to Task 3&4

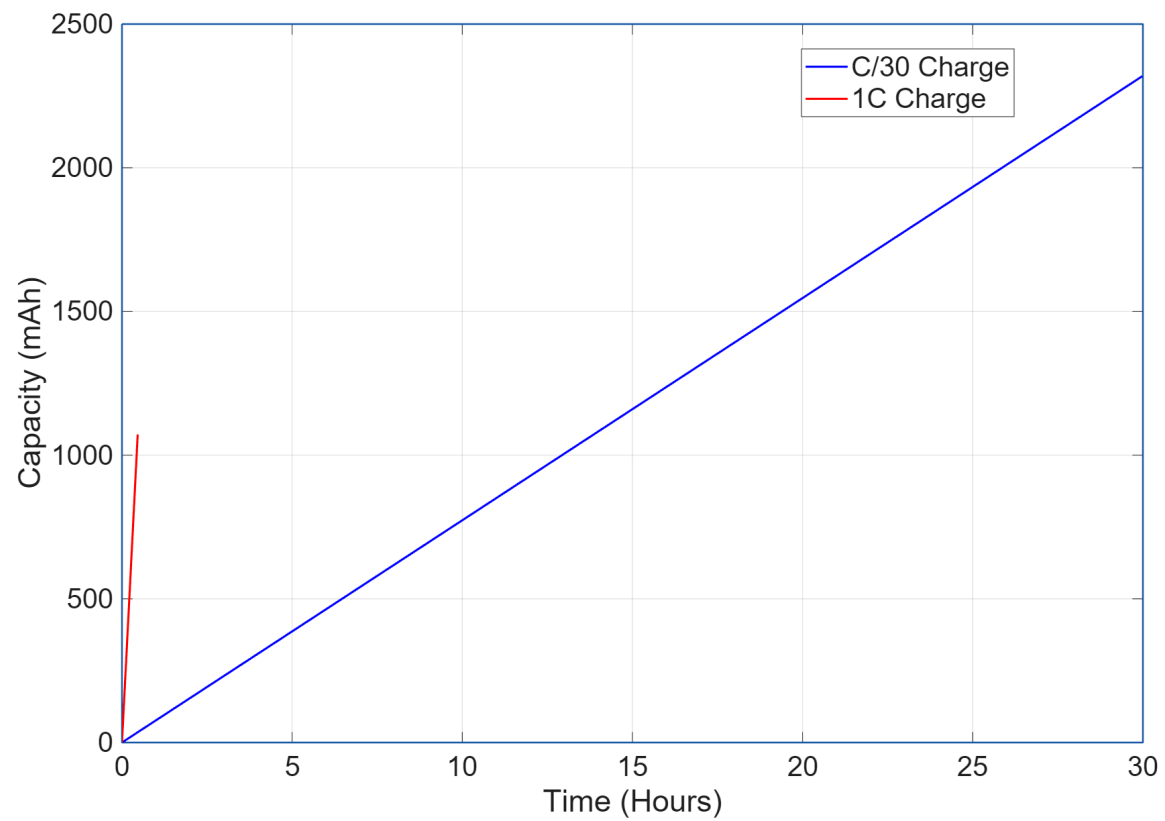


Fig 1.

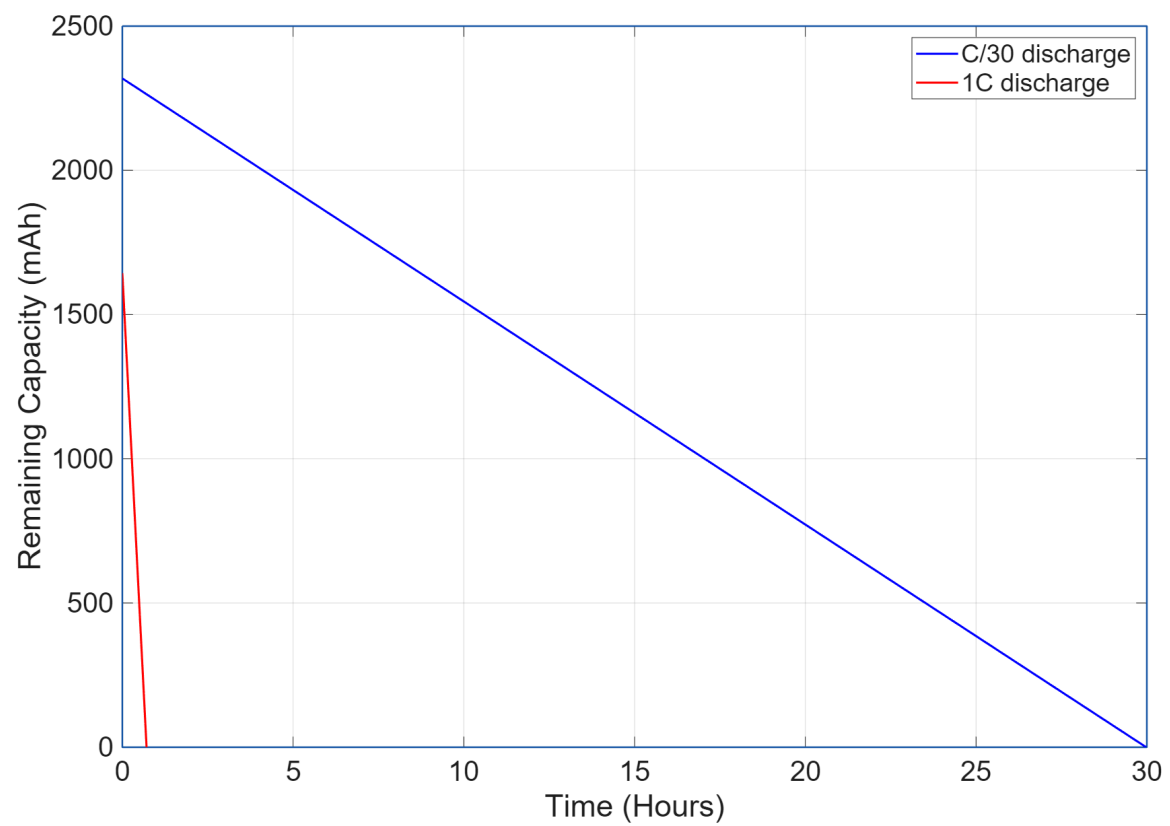


Fig 2.

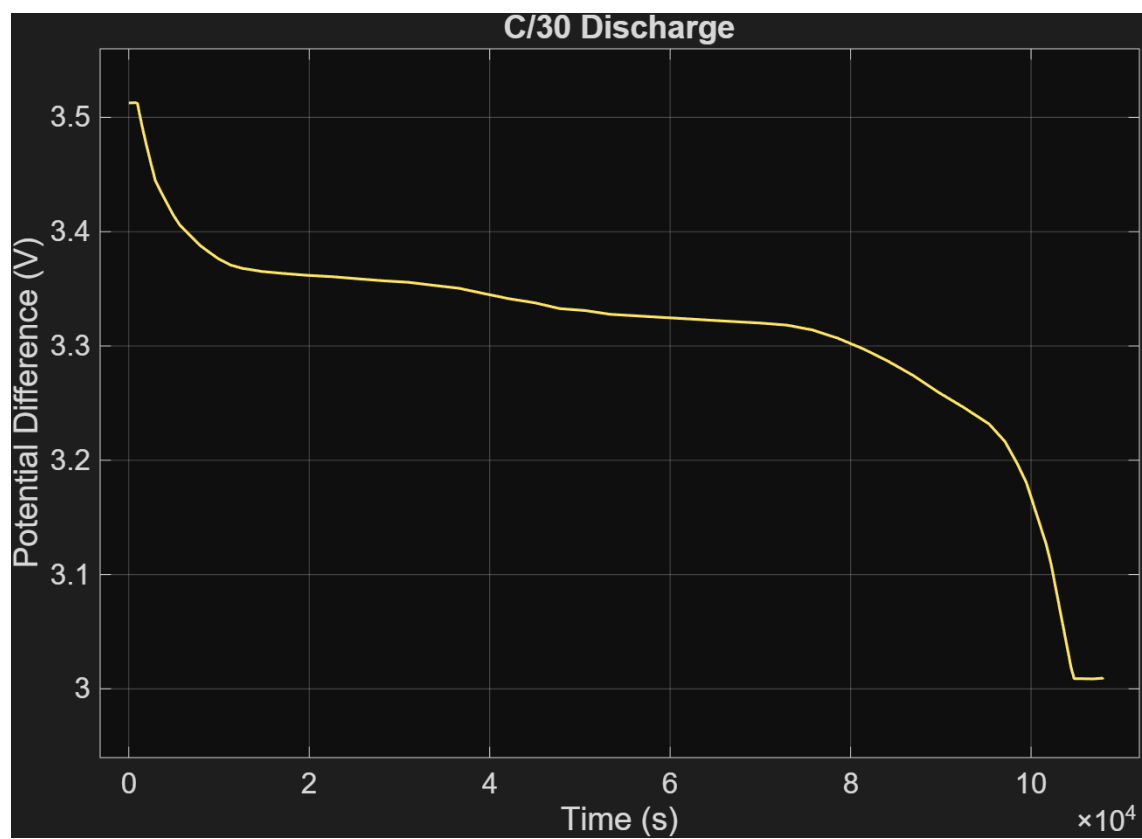


Fig 3.

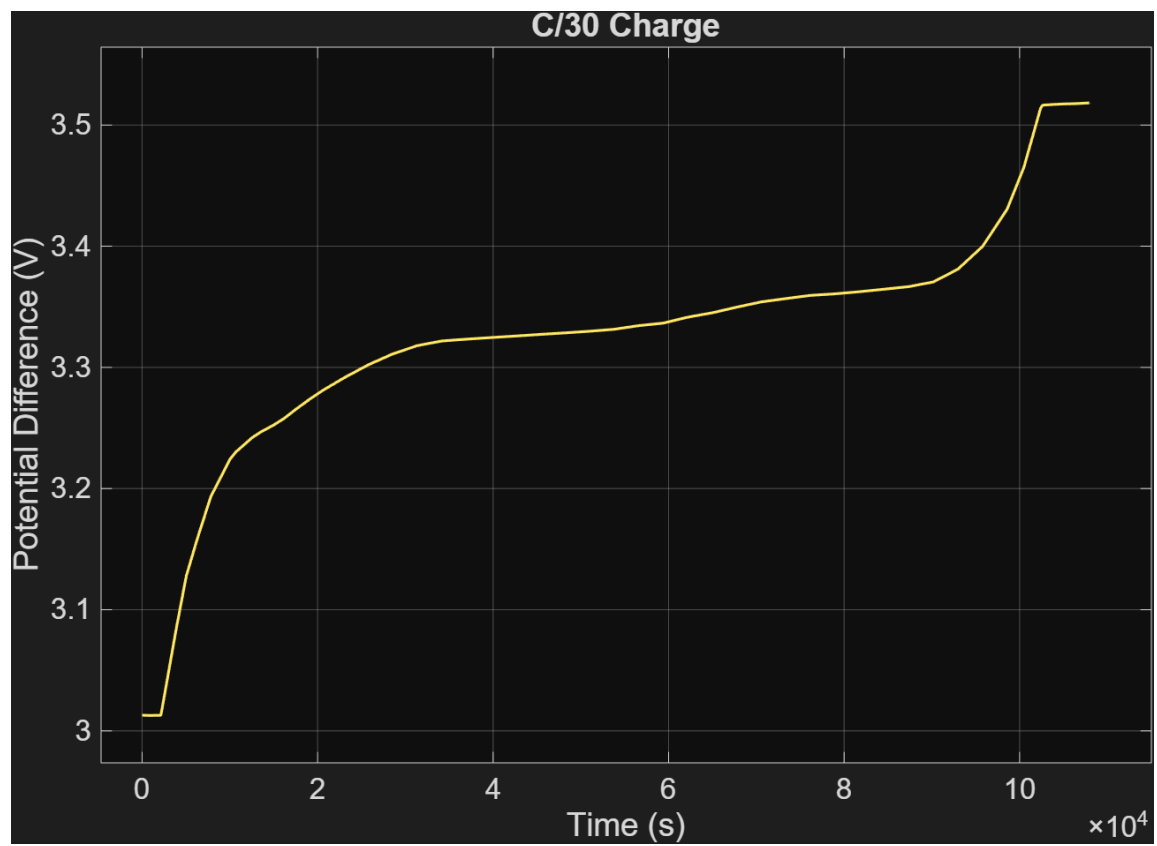


Fig 4.

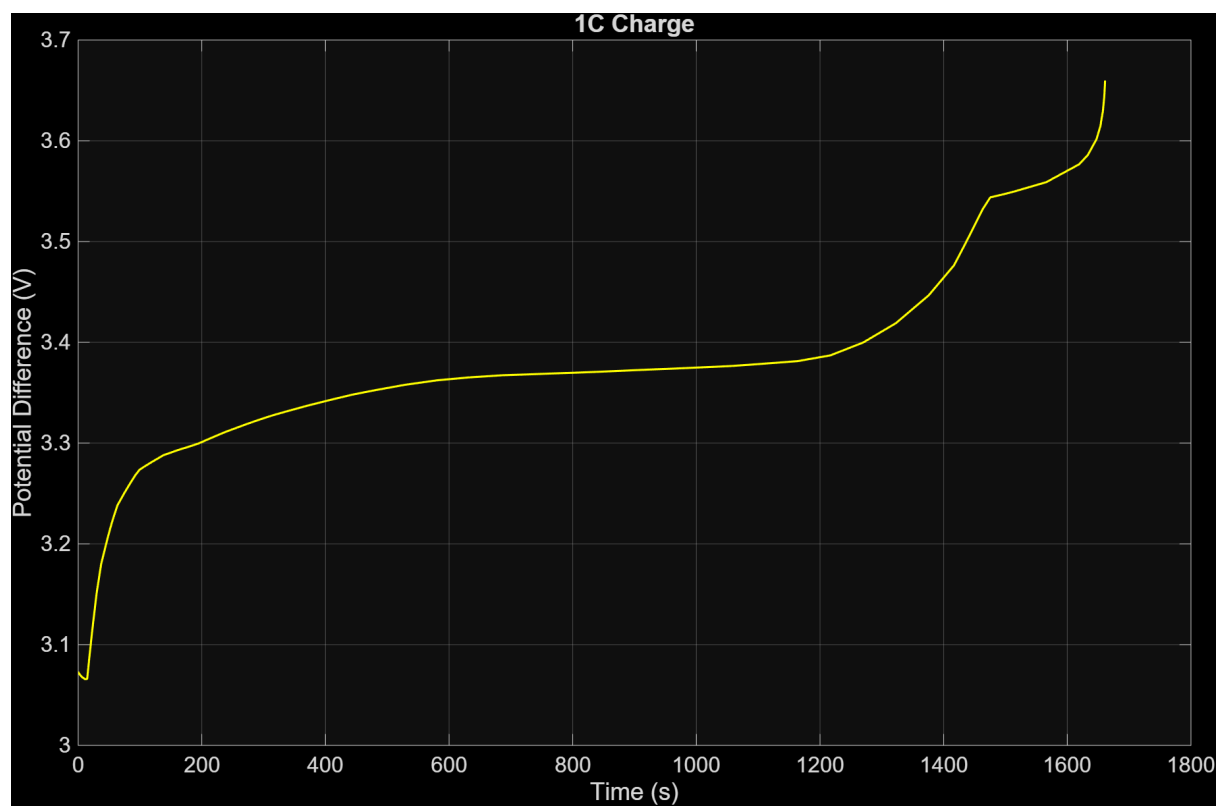


Fig 5.

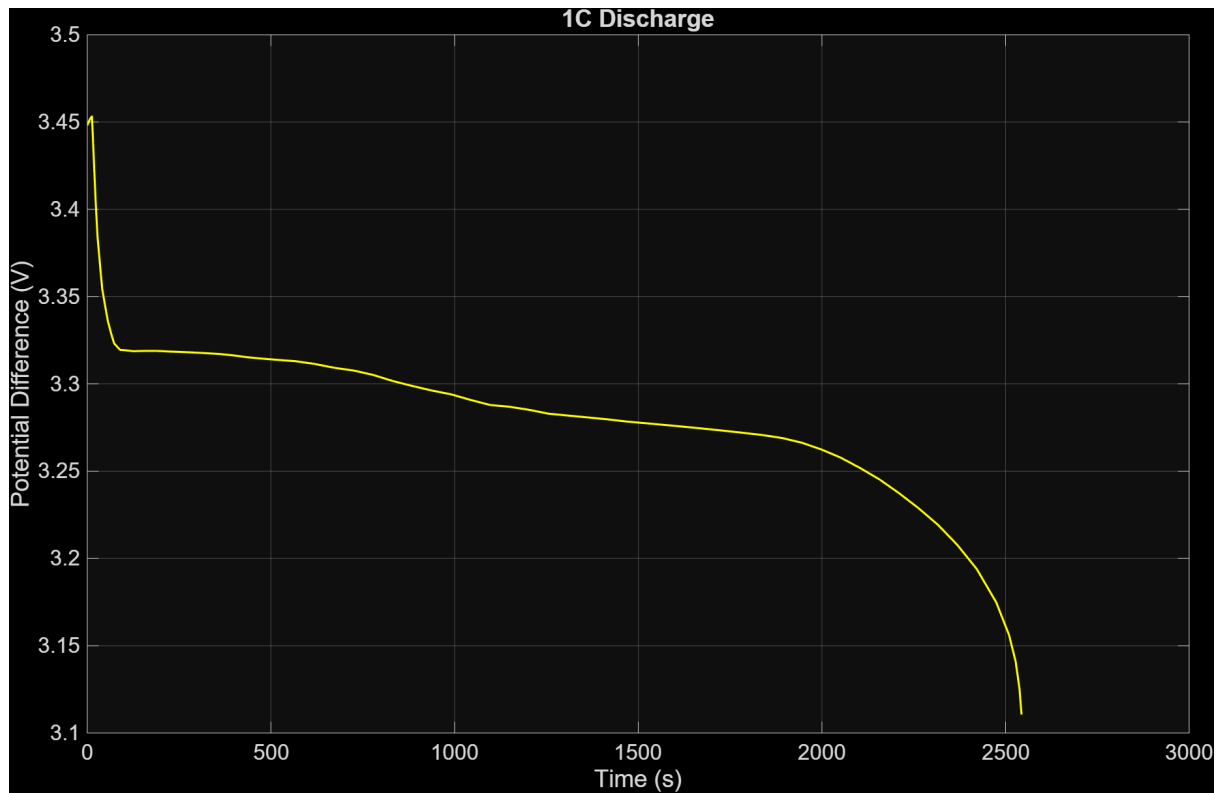


Fig 6.

Answer to Task 5

C/30 case fits to real world scenario. It can be seen that real world testing of another LFP cell in fig 7. has similar voltage vs amper discharge curve. Since simulation is constant current the plot shape is equivalent. Flat voltage profile is due to phase change of cathode.

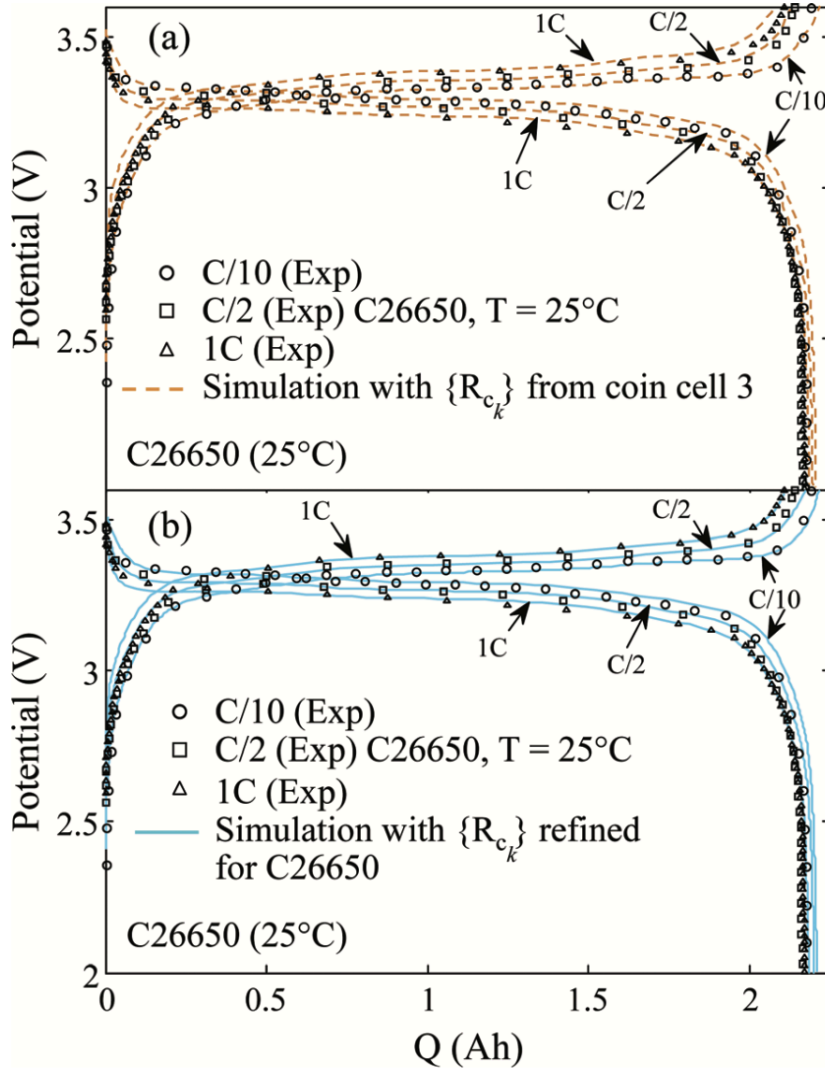


Fig 7. [Modeling of a Commercial Graphite/LiFePO4 Cell](#)

For C/30 discharge and charge scenario, simulated cell capacity Ah matches with the real product both are 2.3Ah.